

# APEKSHA GUPTA

AI Software Engineer — LLM-Powered Backend Services | LLM API Integration, RAG & Vector Search, REST APIs

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LinkedIn | GitHub | Portfolio

## PROFILE

Python engineer with 2+ years of software delivery experience, now building LLM-powered backend services. Hands-on with LLM API integration, prompt design and response handling, vector embeddings in FAISS, and REST API integration with external services (Jira, GitHub). Built and deployed an internal RAG application adopted in production, plus two agentic / RAG systems end to end. Comfortable across the full delivery cycle — design, build, test, deploy and production support — in Agile/Scrum teams. AWS Certified Cloud Practitioner.

## CORE TECHNICAL SKILLS

- LLM & GenAI Engineering:** LLM API integration (model-agnostic — Groq, Hugging Face, Claude), prompt design and response handling, RAG pipelines, vector embeddings in FAISS, LangChain, LangGraph, agentic tool-use / function-calling workflows
- Languages:** Python, Java, SQL
- Backend & APIs:** REST API design and integration, Jira REST API, GitHub REST API / PyGithub, JDBC, Apache Tomcat
- Data & Databases:** SQL-based querying and analysis, Oracle Database, vector databases (FAISS), Pandas, NumPy, Scikit-learn
- Cloud & DevOps:** AWS (EC2, S3, IAM) — Certified, Docker, Git, Linux, Agile SDLC

## AI / ML PROJECTS

### Autonomous AI Dev Agent

Python, LangGraph, Groq LLM API, GitHub REST API (PyGithub), Jira REST API, Streamlit | 2026 | [GitHub Repo](#)

- Integrated the Jira and GitHub REST APIs end to end — the agent reads ticket requirements from Jira, raises pull requests on GitHub and writes status back to Jira, handling authentication, request/response payloads and failure paths for both services.
- Built an autonomous agent that takes a Jira ticket through requirement analysis, code generation, execution, self-correction and PR creation, orchestrated as a 6-node LangGraph state machine with a Streamlit monitoring dashboard.
- Implemented conditional retry and self-correction logic: execution failures are fed back to the LLM through structured prompts for code correction before re-execution.

### Document Q&A Bot — RAG Pipeline

Python, LangChain, FAISS, Hugging Face Embeddings, Groq, Docker, AWS EC2 | 2026 | [GitHub Repo](#)

- Built an end-to-end RAG pipeline — document chunking, Hugging Face embeddings, and vector storage and similarity retrieval in FAISS — with Groq LLM APIs for document-grounded question answering.
- Implemented out-of-scope query handling with relevance thresholds, so the service declines to answer when no relevant context is retrieved.
- Integrated LLM and embedding providers behind a common abstraction layer, keeping the pipeline model-agnostic rather than coupled to one vendor's API; deployed in a Docker container on AWS EC2.

## EXPERIENCE

### AI Software Engineer — Amdocs, Gurugram

Jul 2026 – Present

### Software Engineering Associate — Amdocs, Gurugram

Aug 2024 – Jun 2026

- Designed, built and deployed an internal LangChain + RAG AI agent over migration documentation, adopted by the team in production.
- Built Python automation to extract and analyse application job metadata across hundreds of files (64,000+ lines each) using SQL and scripted parsing, generating structured reports and eliminating hours of manual analysis per cycle.
- Delivered enterprise systems work across two AWS cloud migration initiatives, assessing mainframe and Java applications for cloud readiness and documenting migration findings for stakeholders.
- Investigated and resolved production issues across mainframe-based enterprise applications, performing root-cause analysis across application and data failures.
- Worked the full SDLC in an Agile/Scrum team — design, build, test, deploy and ongoing production support — collaborating with cross-functional stakeholders on requirements and findings.

## EDUCATION

### B.E. Computer Science — Specialization in Big Data Analytics (Hons.), Chandigarh University

2020 – 2024

## CERTIFICATION

### AWS Certified Cloud Practitioner — Amazon Web Services

Feb 2026

## ADDITIONAL INFORMATION

- Target Oct 2026: extending the Document Q&A RAG Bot into a FastAPI microservice — PostgreSQL/pgvector vector store, SQLAlchemy ORM, pytest coverage, Dockerised deployment with GitHub Actions CI.
- Based in Gurugram (Delhi NCR).